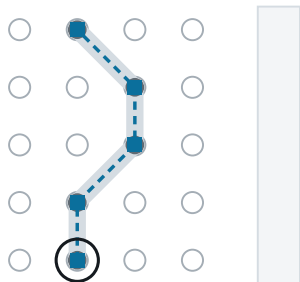


Selection acts on hosts. Which microbial variation can respond?

lineal

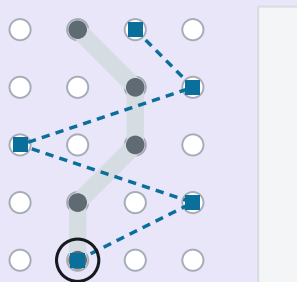
follows the selected hosts'
own line of descent



expected to respond

non-lineal

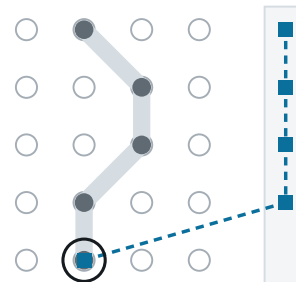
among hosts, but not
these hosts' ancestors



it is not obvious

novel

from outside the
host population



cannot, by definition

grey: the hosts' line of descent; dotted: the ancestry of one of their microbes; shaded column: outside the host population